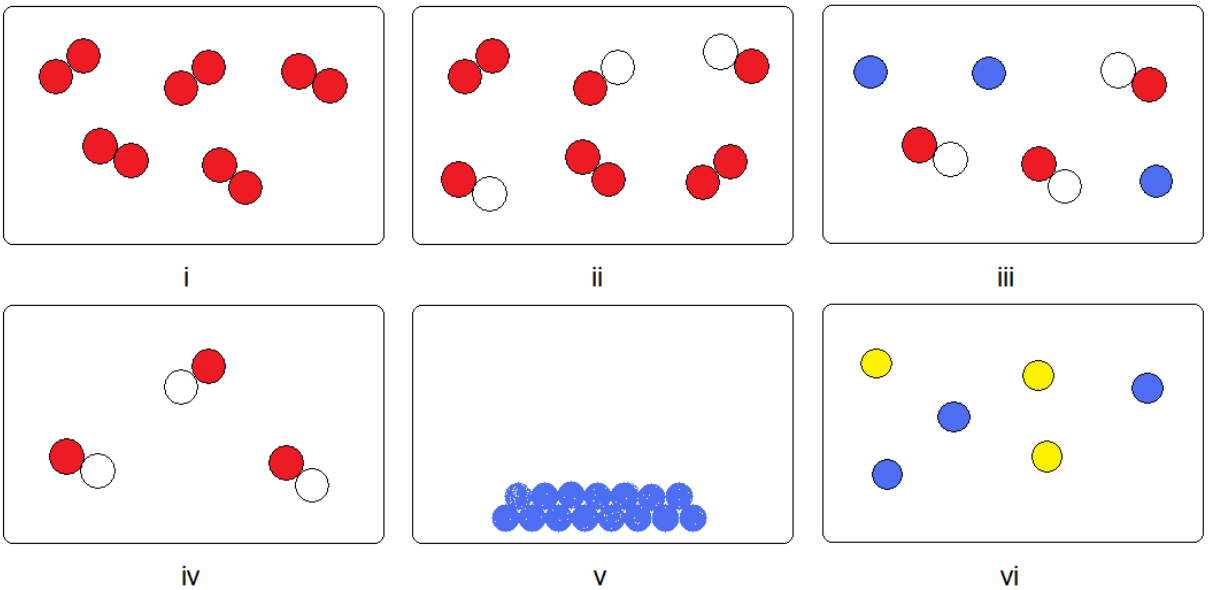
Chem 150 **Answers Problem set 1**

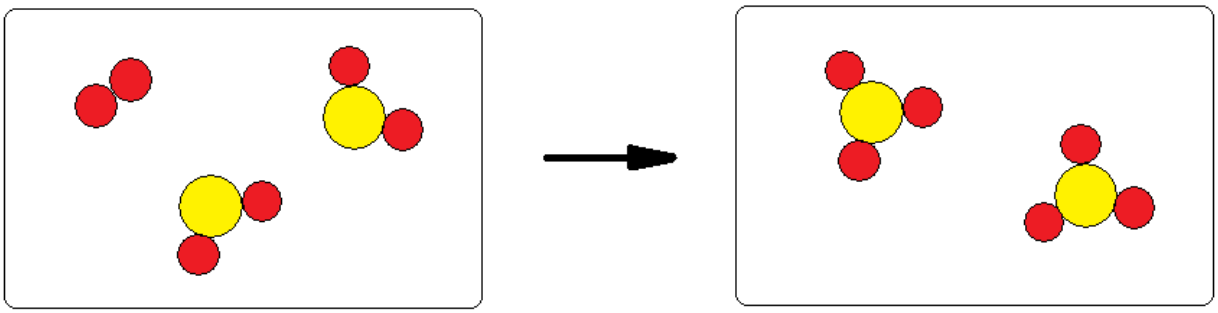
1. Which of the following figures represents?

a) A pure element
b) A mixture of two elements
c) A pure compound
d) A mixture of an element and a compound



i and v: pure element; ii, iii: A mixture of an element and a compound; iv: a pure compound; vi: A mixture of two elements

2. Does the following diagram represent a chemical or physical change? How do you know?



Chemical Change! New matter is produced.

3. Define the following forms of matter as element, compound or mixture. If you indicate mixture specify if homogeneous or heterogeneous:
   a) Sand **heterogeneous mixture**
   b) Diamond **element**
   c) Gasoline **homogeneous mixture**
   d) Sugar **compound**

4. Indicate which of the following changes represent chemical change or physical change:
   a) Burning candle wax chemical change
   b) Melting candle wax **physical change**
   c) Cutting wire **physical change**
   d) Painting your bike **physical change***

   ***Painting in the sense that the paint is made up of pigments, a binder and a solvent. The solvent evaporates and hardens the binder. (There are a few paints out there that chemically react with the surface or with the binder)**

5. Fill in the empty spaces:

Symbol	$^{27}Al^{3+}$	__^{96}Mo	^{150}Sm
# of protons	__13_	42	__62_
# of neutrons	__14_	54	__88_
# of electrons	__10_	__42_	__62_
Charge	__+3_	0	__0__

6. Which atomic property is described by the following equation?

   $Cl + e^- \rightarrow Cl^- + energy$

   Electron Affinity (or electron gain)

7. Name 3 metals, 3 non-metals and 2 metalloids. Name three properties that characterize metals and three properties that characterize non-metals.
   -Metals are ductile, malleable and good conductors of heat and electricity. They also have a shiny appearance (iron, nickel, gold).
   -non-metals are brittle. They are generally not conducting electricity and have no lustrous appearance (bromine, sulfur, argon)
   Examples for metalloids are Silicone, Germanium, Arsenic (note that both aluminum and polonium are metals even though they are bordering the “step ladder”)

8. Balance the following equation

$2C_2H_6O_2 + 5O_2 \rightarrow 4CO_2 + 6H_2O$

$P_4 + 5O_2 \rightarrow P_4O_{10}$

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

$Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$

9. In a typical car battery lead (IV) oxide reacts with lead according to the following equation: (During charging the battery the reveres reaction occurs!)

$PbO_2 + Pb + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$

One way of checking the "livelihood" of a battery is to measure the density of the acid. At first approximation battery acid consists only of sulfuric acid and water. Calculate the amount of lead sulfate that formed and the amount of lead that reacted if the density of the acid changed from 1.2769 g/ml in the freshly charged battery to 1.1243 g/ml. Assume a total of 3 L of battery acid which does not change considerably during the reaction. Use the table densities of aqueous sulfuric acid to solve the problem.

This is a bit of a tricky question. The key to this problem is to realize that sulfuric acid is used up in the discharge of the battery and water is produced. Thus the battery acid is diluted which changes the density. First wee convert densities to specific gravity using the density of water at 4°C which we look up in a handbook or otherwise (0.9999750 g/ml). Next we relate the specific gravity of the sulfric acid to its concentration using the table provided (Excel file).

1.2769 g/ml corresponds to (1.2769 g/ml)/(0.9999750 g/ml) = 1.2769 which gives 37 % H_2SO_4

similarly we obtain for 1.1243 g/ml a concentration of 18 % H_2SO_4

37 % H_2SO_4 refers to 37 g H_2SO_4 in 100 g solution (e.g. battery acid)
We have 3 L of battery acid with a density of 1.2769 g/ml so using D = m/V we get 3830.7 g acid
37 % of this mass is pure sulfuric acid: 0.37 x 3830.7 g = 1417.4 g

For the 18 % acid we get 18 g of H_2SO_4 in 100 g acid. If we assume the volume is not changing (which is a good approximation in this case) we get 3372.9 g acid, 18 % of which is pure sulfuric acid: 0.18 x 3372.9 g = 607.12 g

The difference in mass corresponds to the mass of sulfuric acid used up during battery discharge: 1417.4 g – 607.12 g = **810.28 g**

If we convert this mass of sulfuric acid to moles of sulfuric acids we can relate it to the amount of lead sulfate produced and the amount of lead consumed using the balanced reaction equation.

1.) convert grams to moles

801.28 g H_2SO_4 x (mol/98.08 g) = 8.261 mol H_2SO_4

2.) convert moles to moles

8.261 mol H_2SO_4 x (2 mol $PbSO_4$/2mol H_2SO_4) = 8.261 mol $PbSO_4$

8.261 mol H_2SO_4 x (1 mol Pb/2mol H_2SO_4) = 4.131 mol Pb

3.) convert moles to grams

8.261 mol $PbSO_4$ x (303.26g/mol) = **2505 g $PbSO_4$**

4.131 mol Pb x (207.2 g/mol) = **855.9 g Pb**

answer: During the discharge 2.5 kg of lead sulfate are produced and 0.86 kg of lead is used up.

LEAD Batteries are heavy!!!!

10. Copper can be refined from copper salt solutions using iron according to the following equation

$CuCl_2$ + Fe → Cu + $FeCl_2$

What is the %yield of copper if 150 g Copper chloride reacted and 55 g Copper were obtained.

% Yield = (mass of product actually obtained (55 g Cu) divided by mass of product that could in theory have formed) x 100 %. To find the theoretical yield assume all the copper chloride reacted:
First convert grams of $CuCl_2$ in to mols of $CuCl_2$: 150g x (mol/ 134.45g) = 1.116 mol

Next convert mols of $CuCl_2$ to mols of Cu. This is easy since the molar ratio is 1:1
1.116 mol $CuCl_2$ x (1 mol Cu/1 mol $CuCl_2$) = 1.116 mol Cu

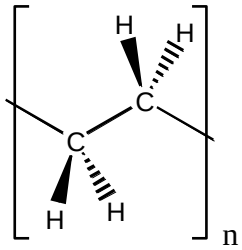
Now convert mols of copper to grams of copper: 1.116 mol Cu x (63.546 g/mol) = 70.9 g Cu

Finally calculate the % yield = {55 g/70.9 g} x 100 % = 77.6 %

11. A main component of Portland cement that is made up of the elements calcium, silicon and oxygen has the following analytical data: 34.50 % Ca, 24.18 % Si. Calculate the empirical formula of the compound.

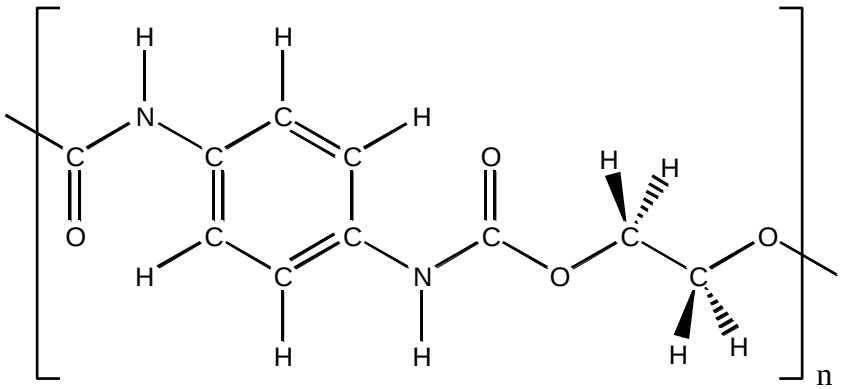
34.50 g Ca x (mol/40.08 g) = 0.8608 mol
24.18 g Si x (mol/28.09 g) = 0.8608 mol
100g – (35.50g + 24.18g) = 41.32 g ; 41.32 g O x (mol/ 16.00g) = 2.5825 mol
Divide the molar ratio by the smallest number (0.8608 mol) and we get a ratio for:
Ca:Si:O = 1:1:3 which gives the empirical formula $CaSiO_3$.

12. You have to determine the identity of a polymer used to insulate a copper wire in an older industrial complex: Elemental analysis of the polymer gave 38.42 % C and 4.88 % H. Which of the following polymers is very likely to be the material in question?



Polyethylene (PE)

Polyethylene can be ruled out immediately since it is made up of the elements hydrogen and carbon only. The %C and %H in the elemental analysis would add up to 100 % which they clearly do not.



Polyurethane (PU)

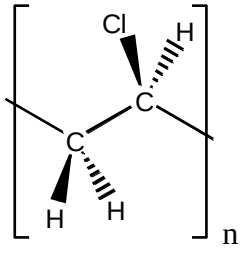
First find the empirical formula of this polyurethane by simply counting all the atoms in the formula unit: C10 H10 N2 O4. This can be reduced to the simplest whole number ratio $C_5H_5NO_2$

Next find the molecular weight (or more accurate the formula weight) of this formula:
(5 x 12.011 g/mol) + (5 x 1.008 g/mol) + 14.01 g/mol + (2 x 16.00 g/mol) = 111.105 g/mol

Determining the % Carbon and % Hydrogen is now a relatively simple task:

{(5 x 12.011 g/mol) /(111.105 g/mol)} x 100 % = 54.05 % Carbon.

Conclusion: This is almost 20 % off. It cannot be this polyurethane!

Polyvinylchloride (PVC) 

By exclusion this should be the material in question. Let's check using the same mechanism as above:
Empirical formula for PVC is C_2H_3Cl which has a mass of:
(2 x 12.011 g/mol) + (3 x 1.008 g/mol) + 35.45 g/mol = 62.496 g/mol

This gives: {(2 x 12.011 g/mol) /(62.496 g/mol)} x 100 % = 38.43 % Carbon
and{(3 x 1.008 g/mol) /(62.496 g/mol)} x 100 % = 4.84 % Hydrogen.

Final conclusion: The elemental analysis are in close agreement with PVC.

13. How much of a 5 M stock solution of NaCl do you need to make 2 L of a 2 M NaCl solution?

$c_iV_i=c_fV_f$
Solving for V_i gives $V_i = c_fV_f/c_i$ and we can substitute V_i = 2M x 2L / 5M = 0.8 L

This means we require 0.8 l of the 5 M stock solution and dilute it with so much water as to end up with 2 L. This new solution will have a concentration of 2M.

14. Name the following ions:

SO_4^{2-}	sulfate (or sulphate)	H_2SO_4	*sulfuric acid (or sulphuric acid)*
Cl^-	chloride	HCl	*hydrochloric acid*
PO_4^{3-}	phosphate	H_3PO_4	*phosphoric acid*
ClO_3^-	chlorate	$HClO_3$	*chloric acid*
NO_3^-	nitrate	HNO_3	*nitric acid*

15. Write names and formulas for the parent acids for the ions above.
To get to the parent acids we add enough hydrogen ions (protons), H^+,to compensate for all the negative charges and obtain the electrically neutral molecules. So a sulfate ion with a charge of minus two will add two hydrogen ions to form sulfuric acid. See *italicised* answer on the right above.